

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Bachelor of Technology (Civil Engineering) SEMESTER - 6 Summer 2025 (Regular)

Course : Bachelor of Technology (Civil Engineering) Branch : Engineering and Technology

Semester : SEMESTER - 6

Subject Code & Name: BTCVC602 - FOUNDATION ENGINEERING

Time : 1.5 hours]

[Total Marks : 60

Instructions to the Students:

1. Each question carries 12 marks.
2. Question No. 1 will be compulsory and include objective-type questions.
3. Candidates are required to attempt any four questions from Question No. 2 to Question No. 6.
4. Use of non-programmable scientific calculators is allowed.
5. Assume suitable data wherever necessary and mention it clearly.

Q1. Objective type questions. (Compulsory Question)

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- 1 Which of the following methods is most suitable for obtaining **undisturbed soil samples**?
A) Split-spoon sampler
B) Open drive sampler
C) Standard penetration test (SPT)
D) Auger boring
- In which of the following tests is an undisturbed sample mandatory?
A) Liquid limit test
B) Triaxial shear test
C) Specific gravity test
D) Sieve analysis
- 3 In the SPT, the split-spoon sampler is driven into the soil using a hammer of weight:
A) 50 kg
B) 63.5 kg
C) 65 kg
D) 75 kg
- 4 Which type of failure is characterized by no bulging at the ground surface and sudden punching of the footing into the soil?
A) General shear failure
B) Local shear failure
C) Punching shear failure
D) Lateral shear failure
- Safe bearing capacity is defined as:
A) Load causing elastic deformation
B) Load at which failure occurs
C) Ultimate capacity divided by factor of safety
D) Pressure from wind loads
- 6 Terzaghi's bearing capacity equation is applicable for:
A) Deep foundations
B) Cohesive soils only
C) Shallow strip footings
D) Raft foundations

- 7 Which type of footing is recommended when the soil bearing capacity is very low?
 A) Isolated footing
 B) Combined footing
 C) Raft (Mat) footing
 D) Trench footing
- 8 A common clay mineral responsible for swelling behavior in expansive soils is:
 A) Kaolinite
 B) Illite
 C) Montmorillonite
 D) Chlorite
- Negative skin friction occurs when:
 A) Ground settles faster than the pile
 B) Pile settles faster than the soil
 C) Soil is fully consolidated
 D) Pile is too short
- Sheet piles are primarily used to:
 A) Increase bearing capacity
 B) Provide vertical support for buildings
 C) Retain soil and resist lateral pressure
 D) Support bridge decks
- 11 The Factor of Safety (FOS) in slope stability is defined as:
 A) Ratio of driving forces to resisting forces
 B) Ratio of resisting forces to driving forces
 C) Difference between resisting and driving forces
 D) Sum of driving and resisting forces
- The stability number in slope stability refers to:
 A) A dimensionless number indicating slope stability for given soil parameters
 B) Soil moisture content
 C) Slope angle in degrees
 D) Total load on the slope
- Q2. Solve the following.
 A) What are the purposes of soil exploration? 6
 B) Explain the Electric Resistivity Method? 6
- Q3. Solve the following.
 A) State and explain the below terms 6
 1) Net Pressure Intensity,
 2) Net Ultimate bearing capacity, and
 3) Net Safe bearing capacity
 Explain the terms in Terzaghi's bearing capacity eq. developed using general shear failure? 6
- Q4. Solve Any Two of the following.
 What are expansive soils? Explain in detail the characteristics of expansive soils? 6
 Define and explain the following terms: 6
 1) Free Swell
 2) Swelling Potential
 3) Swelling Pressure
 C) Define settlement of foundation. Explain the different types of settlement in shallow foundations. 6
- Q5. Solve Any Two of the following.

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- Q6. Give Any Two of the following.
- A) A group of 9 piles, with 3 piles in a row, was driven into soft clay extending from ground level to a great depth. The diameter and length of piles were 30 cm and 10 m, respectively. The unconfined compressive strength of clay is 70 kN/m^2 . If the piles were spaced at 90 cm c/ c, compute the allowable load on the pile group on the basis of shear failure criteria for a factor of safety of 2.5. Neglect bearing at the tip of the piles. Take $m = 0.6$ for shear mobilization around each pile. 6
- B) Explain the sheet pile walls in detail using neat sketches. 6
- C) State and explain in detail the Engineering News Record Formula (ENR) and Hiley's Formula? 6
- A) What are the different types of slopes. Explain in detail with neat sketch and example? 6
- B) A slope is to be constructed at an inclination of 30° with the horizontal. Determine the safe height of the slope at factor of safety of 1.5. The soil has the following properties: $c = 15 \text{ kN/m}^2$, $\phi = 22.5^\circ$ and $\gamma = 19 \text{ kN/m}^3$. 6
- C) Enlist and explain in detail the different types of slope failures? 6

*** End ***

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